Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps
Subject In-charge	Prof. Snehal Andhare

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Roll Number	23102A0029
Grade and Subject Teacher's Signature	

Experiment Number	06	
Experiment Title	To Setup and Run Selenium Tests in Jenkins Using Maven.	
Resources / Apparatus Required	Hardware: Computer system	Software: Intellij, Tomcat, jenkins

Program

Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>SeleniumJenkinsDemo</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <dependencies>

        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>4.35.0</version>
```

```
    </dependency>

    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>5.13.4</version>
      <scope>test</scope>
    </dependency>

  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-surefire-plugin</artifactId>
        <version>3.5.3</version>
      </plugin>
    </plugins>
  </build>

</project>
```

ExampleTest.java:

```
package tests;

import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;

import static org.junit.jupiter.api.Assertions.assertEquals;

public class ExampleTest {

  WebDriver driver;

  @BeforeEach
  public void setUp() {
    ChromeOptions options = new ChromeOptions();

    // Important for Jenkins
    options.addArguments("--headless=new");

    // Browser size in headless mode
    options.addArguments("--window-size=1920,1080");
```

```
    driver = new ChromeDriver(options);
  }

  @Test
  public void verifyExampleWebsiteTitle() {
    driver.get("https://example.com");

    String actualTitle = driver.getTitle();

    System.out.println("Page Title = " + actualTitle);

    assertEquals("Example Domain", actualTitle);
  }

  @AfterEach
  public void tearDown() {
    if (driver != null) {
      driver.quit();
    }
  }
}
```

Output

Selenium using Maven first:

```
C:\Users\23102B0006\Documents\SeleniumJenkinsDemo>mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:SeleniumJenkinsDemo >--------------------
[INFO] Building SeleniumJenkinsDemo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ SeleniumJenkinsDemo ---
[INFO] Deleting C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ SeleniumJenkinsDemo ---
[INFO] skip non existing resourceDirectory C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ SeleniumJenkinsDemo ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ SeleniumJenkinsDemo ---
[INFO] skip non existing resourceDirectory C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ SeleniumJenkinsDemo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.3:test (default-test) @ SeleniumJenkinsDemo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running tests.ExampleTest
Sept 01, 2026 12:24:28 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 151
Sept 01, 2026 12:24:28 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 151.0.7922.174. You may need to include a dependency on a specif
:4.35.0` where the version ("v86") matches the version of the chromium-based browser you're using and the version
Page Title = Example Domain
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.280 s -- in tests.ExampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.222 s
[INFO] Finished at: 2026-09-01T12:24:29+05:30
```

Configure Java and Maven inside Jenkins:

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish JUnit test result report

Test report XMLs

Fileset 'includes' setting that specifies the generated raw XML report files, such as 'myproject/target/test-reports/*.xml'. Basedir of the fileset is the workspace root.

target/surefire-reports/*.xml

Test output retention

None

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

```
cd C:\Users\23102B0006\Documents\SeleniumJenkinsDemo
mvn clean test
```

Advanced

Execute Windows batch command

Command

See the list of available environment variables

```
mkdir "%WORKSPACE%\target\surefire-reports"
copy "C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\target\surefire-reports\*.xml" "%WORKSPACE%\target\surefire-reports\"
```

Advanced

+ Add build step

First Jenkins Selenium job:

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/platform/junit-platform-launcher/1.13.4/junit-platform-launcher-1.13.4.jar (223 kB at 3.9 MB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running tests.ExampleTest
Sept 01, 2026 12:15:56 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 151
Sept 01, 2026 12:15:56 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 151.0.7922.174. You may need to include a dependency on a specific version of the CDP using something similar to
`org.seleniumhq.selenium:selenium-devtools-v86:4.35.0` where the version ("v86") matches the version of the chromium-based browser you're using and the version number of the artifact is the
same as Selenium's.
Page Title - Example Domain
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.557 s -- in tests.ExampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  25.238 s
[INFO] Finished at: 2026-09-01T12:15:57+05:30
[INFO] ------------------------------------------------------------------------
Finished: SUCCESS
```

Publish Selenium test results in Jenkins:

```
[INFO] ------------------< com.example:SeleniumJenkinsDemo >-------------------
[INFO] Building SeleniumJenkinsDemo 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ SeleniumJenkinsDemo ---
[INFO] Deleting C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ SeleniumJenkinsDemo ---
[INFO] skip non existing resourceDirectory C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ SeleniumJenkinsDemo ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ SeleniumJenkinsDemo ---
[INFO] skip non existing resourceDirectory C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ SeleniumJenkinsDemo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.3:test (default-test) @ SeleniumJenkinsDemo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running tests.ExampleTest
Sept 01, 2026 12:19:24 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 151
Sept 01, 2026 12:19:24 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 151.0.7922.174. You may need to include a dependency on a specific version of the CDP using something similar to
`org.seleniumhq.selenium:selenium-devtools-v86:4.35.0` where the version ("v86") matches the version of the chromium-based browser you're using and the version number of the artifact is the
same as Selenium's.
Page Title = Example Domain
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.816 s -- in tests.ExampleTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.460 s
[INFO] Finished at: 2026-09-01T12:19:24+05:30
[INFO] ------------------------------------------------------------------------
[Selenium-Maven-Test] $ cmd /c call C:\Windows\TEMP\jenkins16086282947651738914.bat

C:\ProgramData\Jenkins\.jenkins\workspace\Selenium-Maven-Test>mkdir "C:\ProgramData\Jenkins\.jenkins\workspace\Selenium-Maven-Test\target\surefire-reports"

C:\ProgramData\Jenkins\.jenkins\workspace\Selenium-Maven-Test>copy "C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\target\surefire-reports\*.xml"
"C:\ProgramData\Jenkins\.jenkins\workspace\Selenium-Maven-Test\target\surefire-reports\"
C:\Users\23102B0006\Documents\SeleniumJenkinsDemo\target\surefire-reports\TEST-tests.ExampleTest.xml
       1 file(s) copied.

C:\ProgramData\Jenkins\.jenkins\workspace\Selenium-Maven-Test>exit 0
Recording test results
[Checks API] No suitable checks publisher found.
Finished: SUCCESS
```



Conclusion:	Selenium tests were successfully automated using Maven and executed in Jenkins, with JUnit test results published successfully.